

A summary of Satellite Imaging

Before the details of this project are revealed, let us revisit exactly what this project is. There are hundreds of thousands of mines across the continental United States: tungsten, gold, coal, etc. The central hypothesis of this critical materials hub is that each of these mines have waste dumps, and that these waste dumps may contain both highly abundant and accessible amounts of those aforementioned critical materials. However, before venturing out upon that endeavor, a hurdle must be overcome. Although locations and rough polygons for thousands of mine features (chief among which are the waste dumps) are available, the volumes of many of these sites are genuinely a mystery. If these volumes were to be found, coupled with an estimate of critical materials in parts per million, it would allow prospective extractors to gain an idea of the value of the entire mine dump.

This project delivers one term in that valuation. The value of a dump is, roughly, $Value = Volume \times Grade \times Price \times Recovery$ — cubic metres of waste, times critical-mineral concentration (ppm/grade), times commodity price, times the fraction economically extractable. Grade comes from assays; price from markets; recovery from metallurgy. **Volume is the term that is missing at scale**, because while the inventory gives footprints, it rarely gives depths. Everything that follows is an attempt to supply that one number — volume — from remote elevation data alone, so that a prospector holding a grade estimate can convert a map polygon into a defensible tonnage.

Step Zero / One

This project, Satellite Imaging, sets out to fill that information gap. The resources we used are the following, and for future reference the following will be referred:

- Waste_Geology.xlsx as Geology
- Waste_Resources.xlsx as Resources
- USGS_TopoMineSymbols_24k_Polygons.CPG,DBF,PRJ,SHP,SHX as Topo
- Waste_Polygons.CPG,DBF,PRJ,SHP,SHX as Polygons

We now have appropriate context to understand the project. Let us therefore begin with step one. At this point, we have two large polygon-based datasets (Topo and Polygons) with each marking the coordinates and rough shapes of the mine features, marked with FTR_ID. Geology and Resources allow us to filter out to features that have a volume well

documented; overlap between these features as well as the best footprints off of Topo and Polygons would represent our validation dataset.

The first step is to find the best possible sites to build an algorithm around: sites that are very large, with easily distinguishable footprints, and a flat surrounding terrain. Taking a step back, it is important to also sample from areas with many sites, to enforce defensible conclusions with statistically significant results. In the Jupyter notebook `Mine_Waste_Documentation` (moving forward referred to as step zero), with the module `query_USGS_geospatial_v2`, we clearly see where statistically defensible results will be found:

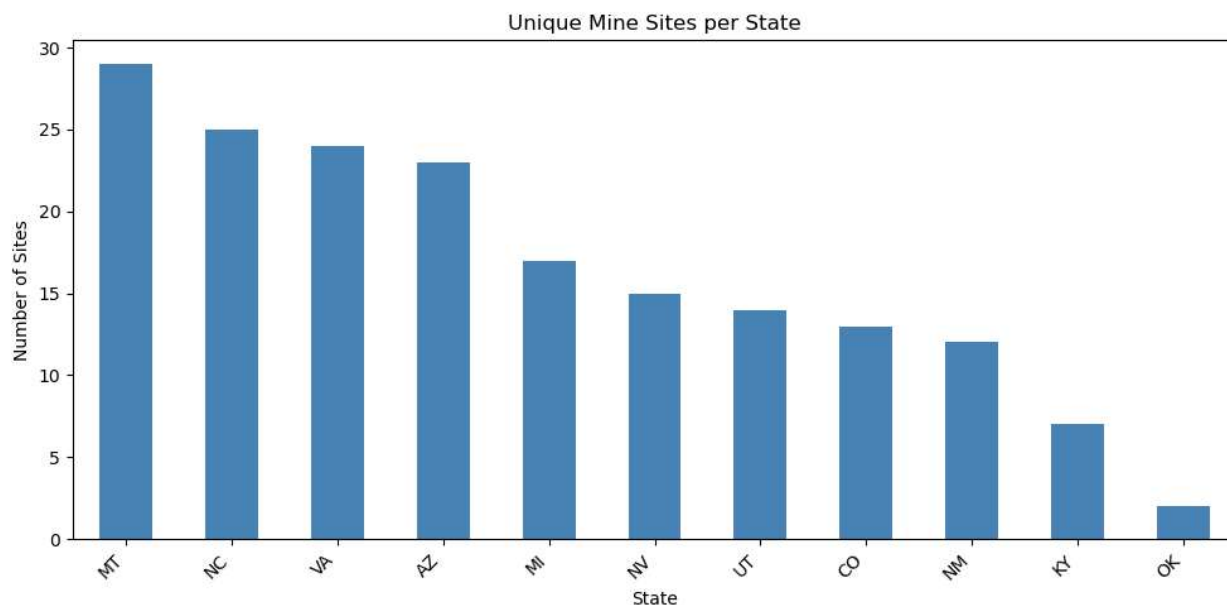


Figure One: Mine Sites in Resources arranged by state.

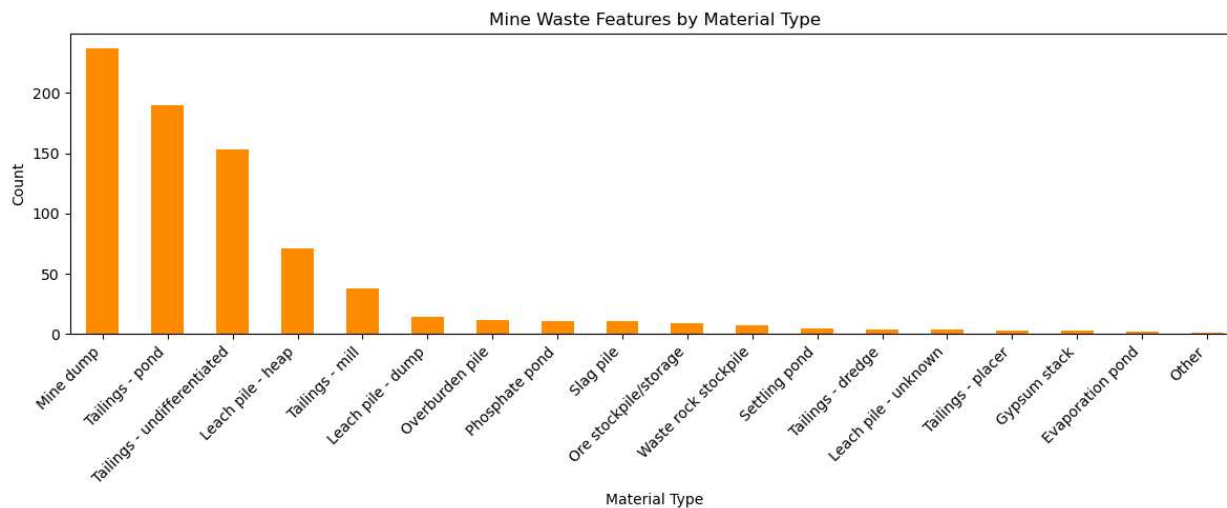


Figure Two: Mine Features in Resources arranged by feature.

Figure one shows that Montana, North Carolina, Virginia, and Arizona represent the states with the most sites, and so our attention should focus on them. Within those states, we should look at mine dumps as well as tailings. However, our focus can be notably narrowed to better sites. Arizona, as well as North Carolina, offers significantly less terrain challenge compared to the mountainous regions of Virginia and Montana. Coupled with that fact, tailings in general are more difficult to calculate volume than mine dumps, with tailings ponds specifically being a negative elevation relative to their surroundings. For these reasons, the focus of the project was kept to Mine dumps in North Carolina, and Arizona.

The results of step zero are reflected in notebook Step_One_Documentation, which uses modules Step1_NC_MineDump_Export_cells, Step1_NC_MineDump_Export, and Step1_AZ_MineDump_Export_cells to export to several .csv and .xlsx files that the next steps will reference. In these files will be found the best 5 sites from AZ and NC, which represent the algorithm's best starting points. Thus concludes Step Zero/One.

Step Two

At this point, the resources being used are: Polygons, Topo, Resources, Top Five NC Sites (see step one), and Top Five AZ Sites (see step one). With this information and an idea of a basic algorithm, a program is written: Step2_BoundingBoxFinder.py. This is imported as a module into Step_Two_Documentation.ipynb, where it serves as the designator of coordinates around mine features. These coordinates form a square, and these bounding boxes are used to import a central aspect of our project: Digital Elevation Models, or DEM's. DEM's are an incredible visualization tool; they are three-dimensional maps of the ground surface, derived from airborne lidar and photogrammetry rather than from satellite photographs, allowing one to capture the elevation of a particular landscape.

The notebook first uses the imported bounding box finder for each of the ten sites – five from AZ and NC, respectively – and padding on all sides about 20%. From there, it pulls DEM data. Notably, because DEM data is storage intensive, the function is only able to remotely download and use pixels 10 meters by 10 meters. This is not nearly as precise as desired, but for now, it will have to do. Each pixel, 100 square meters in size, is a flat square which has the median elevation of the ground in that pixel.

The difference of slope between the pixels is also calculated in a gradient function. Finally, the polygon layer itself would be overlaid. Three panels would be rendered, one with just the raw satellite image, as well as a slope gradient, and the all-important DEM

data. This would be looped over all 10 sites assessed, and the program would write DEM and slope rasters to disk as .tif files for use.

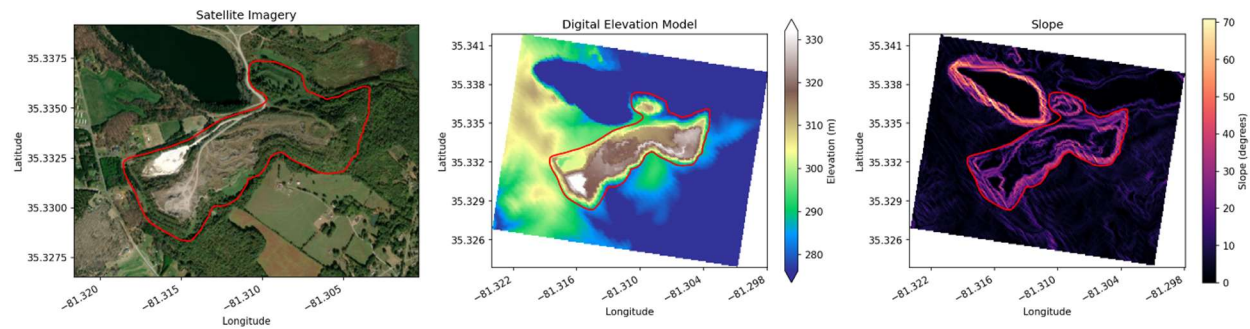


Figure Three: One of the more successful attempts – Hallman Beam, NC

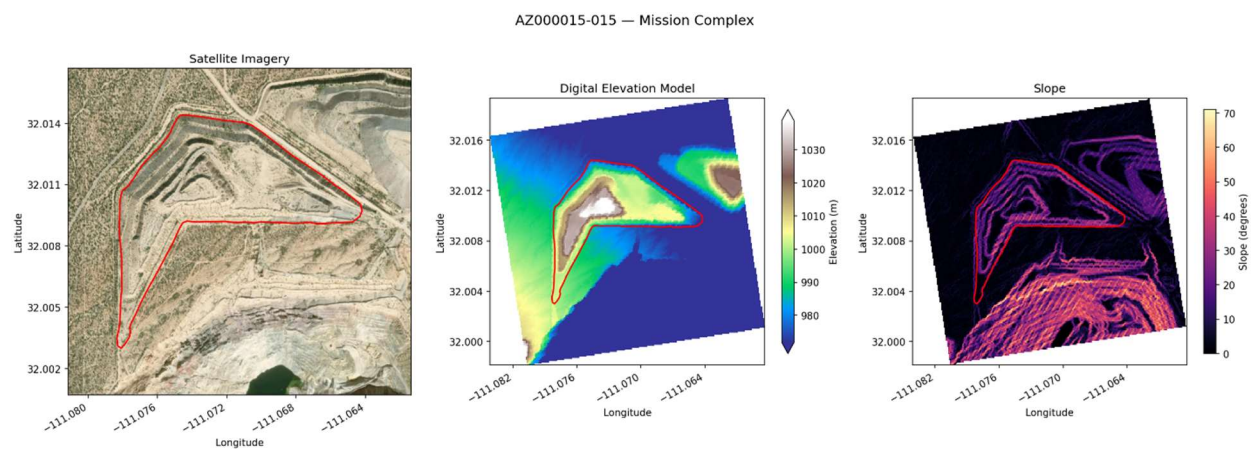


Figure Four: Another successful attempt – Mission Complex

Step Three

Once again, we have Polygons, Topo, Resources, the previous step's saved outputs, and some notes to start us off. This notebook, `Step_Three_Documentation.ipynb`, first pulls two things. The previously found DEMs are used, as well as a new object. From Topo, not only the polygon of the mine waste feature is pulled, but also a polygon of the entire mine site. To find the volume of the mine waste, we need two things: the exterior shape of the feature (provided by DEM), and the ground beneath it. Here is where problems begin.

Obviously, without previously drawn maps of the mine before mining began, it is impossible to know what the ground looks like using satellite imaging. Therefore, it must be extrapolated using the surrounding exposed surface. Subtracting the exterior from the extrapolated ground would theoretically provide the volume of the site.

The disturbance mask, as previously mentioned, would not be used to extrapolate; it is not natural ground. Instead, the bounding box would be enlarged, and a buffer created around the disturbance mask to provide adequate data for extrapolation. To ensure the extrapolation algorithm is actually able to accurately recreate the ground, a series of validation squares are also employed. These squares survey a 300 by 300 meter encampment. One of the squares is relatively natural terrain; median slope across the 9 DEM cells it bounds, one relatively steep, and one relatively flat. The validation squares reveal that the algorithm is adequately successful at recreating ground from nearby data.

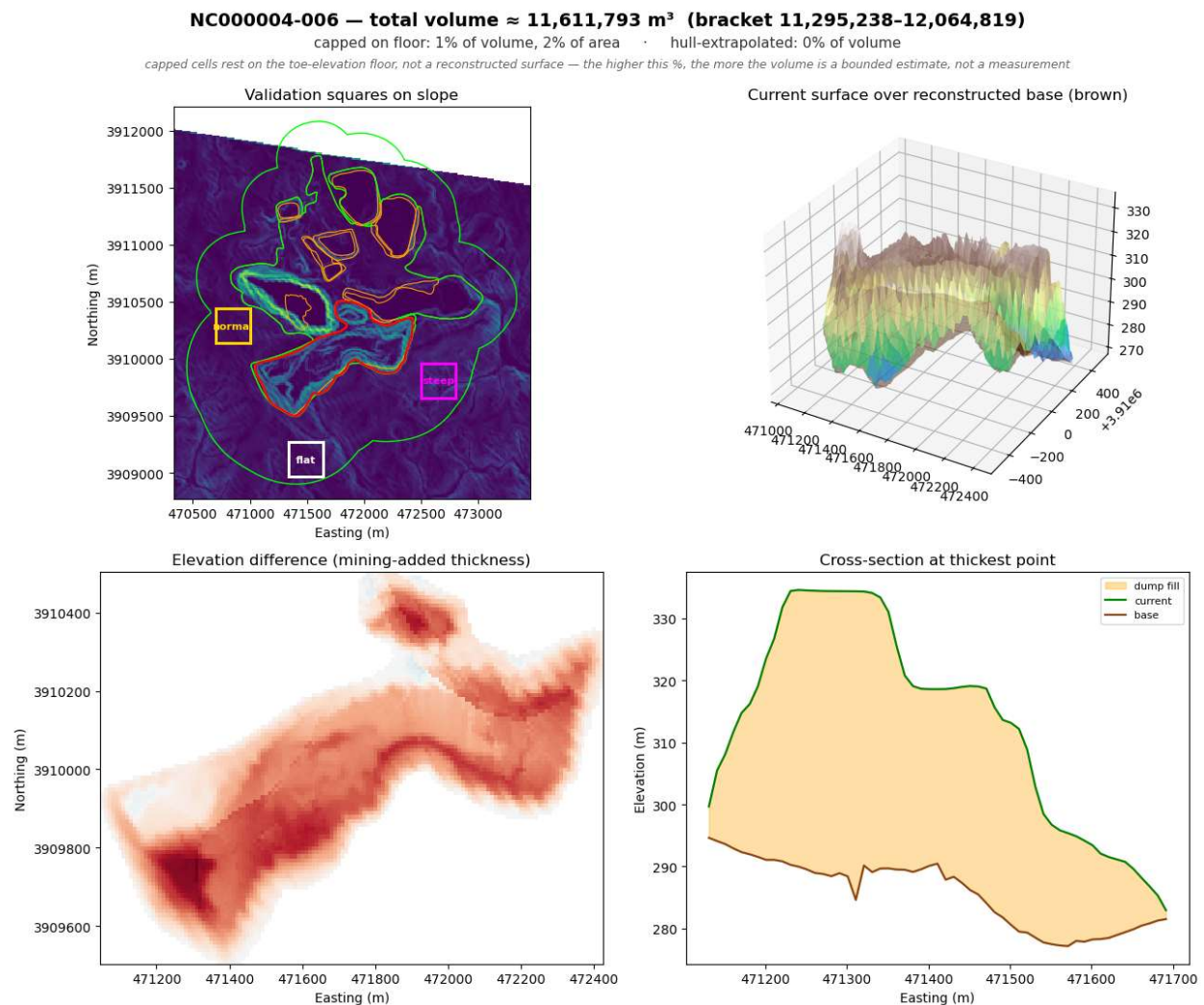


Figure Five: A render of the Hallman-Beam site as it was processed. The green lines indicate the buffer for the extrapolation, the squares revealing the validation, the mask in orange. Directly to the left is the DEM data overlayed over the extrapolated ground, which is on the bottom right.

The main idea behind the thin-plate-spline, or TPS, method used to extrapolate the ground is not complex. Thin Plate Spline (TPS) interpolation is a mathematical method for fitting a perfectly smooth surface across a set of scattered data points. It gets its name from a physical analogy: it calculates the exact shape a flexible, infinitely thin sheet of metal would take if it were pinned to specific heights at various data points and left to bend with minimal resistance. In its pure interpolation form, the resulting surface passes exactly through every single control point provided, ensuring no data loss at the source locations.

Results, however, were less than promising. Both Hallman-Beam and Mission Complex failed to achieve volume numbers similar to the USGS numbers in Resources. At this early two-site stage (the v2 pipeline) Hallman-Beam carried an error of roughly -14%, analyzed in detail below; Mission Complex likewise fell short. As the pipeline matured and was run across all 20 sites in Step Four, the finalized batch figure for Hallman-Beam NC000004-006 settled at -22.9% – est 11,611,793 m³ – a difference reflecting later pipeline refinements, not a change in the underlying conclusion. The rest of the project would strive to correct these errors.

Specifically for Hallman-Beam, the results are clear. We built two fixes: a wider sampling buffer and explicit validation. Neither could close the -14 % gap to USGS, and that null result is the finding. The buffer was not the problem. Going from a 300 m ring (v1) to a 600 m ring + 300 m topo collar (v2) moved the estimate from 12,935,194 m³ (-14.1 %) to 12,988,490 m³ (-13.8 %) – about 0.3 points. The gap is robust to how much surrounding terrain the spline sees.

The reconstruction is accurate, not suspect. Held-out validation squares give bias -0.11 m (flat), -0.19 m (normal), -0.20 m (steep), with RMSE 0.26 / 0.45 / 0.80 m. The spline reproduces undisturbed ground to within ~0.2 m even on the steepest natural patch available. The four-estimate bracket (original + flat/normal/steep) is 12.87–12.99 M m³, only ~0.9 % wide, and sits entirely ~14 % below USGS's 15.06 M, which is far outside it. The biases are *negative* (the base is predicted slightly too low, so volume is slightly over-counted), so correcting for them moves the estimate *down**, away from USGS. We cannot blame the gap on a base fitting too high.

To reach USGS's 15.06 M, the buried base would have to average about 3.4 meters deeper than our reconstruction across the 604,000 m² footprint. Our validated

reconstruction error is roughly 0.2 meters, so a 3.4 m systematic offset is $\sim 17\times$ our measured error; this is to state definitively that it cannot be interpolation error.

The -14% is real, systematic, and not caused by our sampling extent or interpolation. By elimination then, an inference from ruling other causes out, not a directly tested result, the most likely remaining explanation is a base-surface definition difference. USGS's calculated volume is computed against a base nearly 4 meters lower than the surrounding undisturbed terrain projects, by a method left undocumented – and/or a buried excavation or valley deeper than the perimeter implies. The latter is the one cause this method cannot test: surface terrain cannot reveal a void it does not touch.

Caveats:

- (1) The leave-out validation certifies accuracy on natural ground over ~ 100 m gaps; it does not certify the ~ 600 m+ buried gap under the dump.
- (2) The three validation squares landed clustered in one south-east pocket (the steepest fully-natural 100 m patch was only moderately steep), so they probe local accuracy in one location, not across the AOI; spreading them across sectors would strengthen the claim.

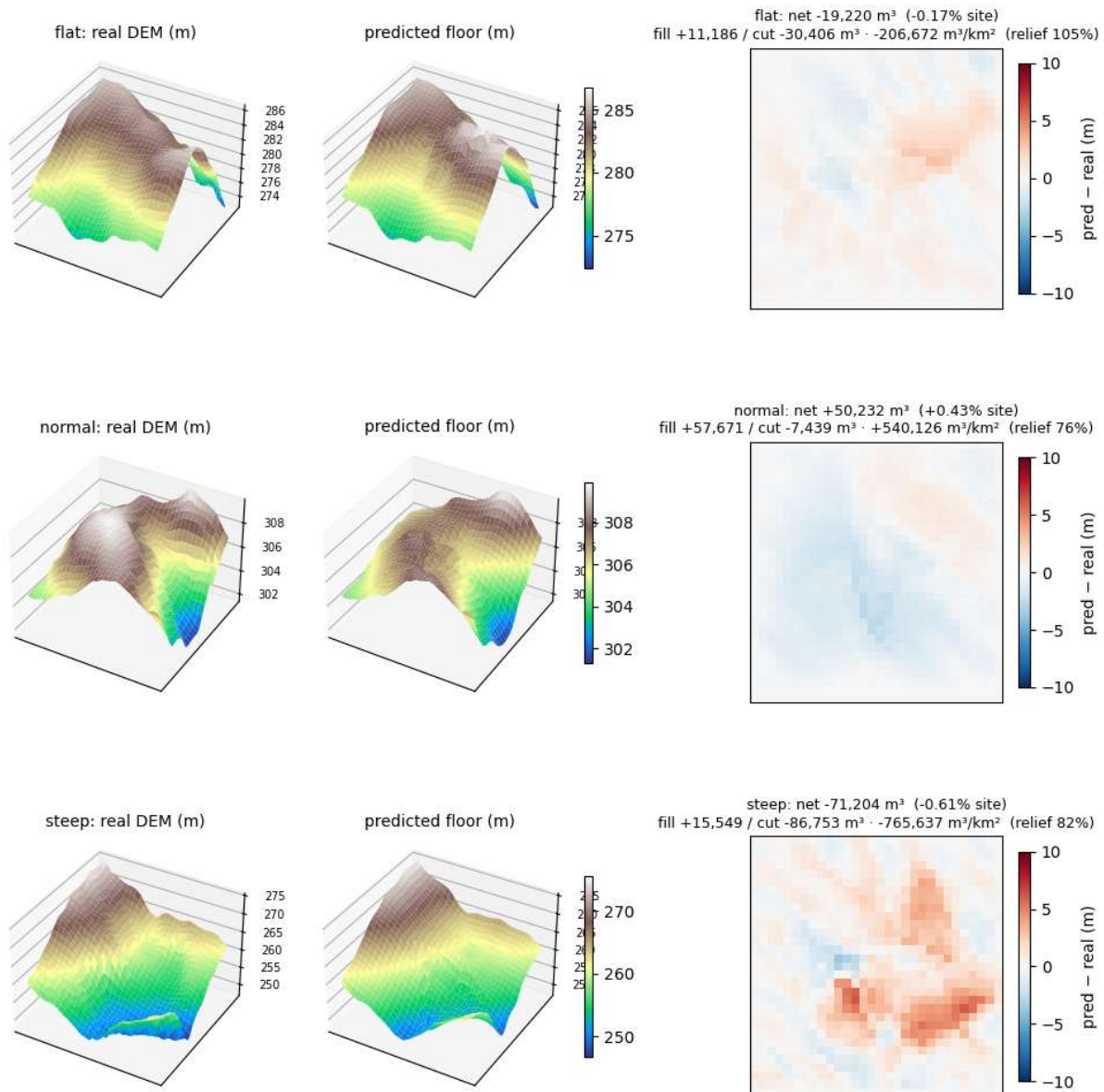


Figure 6: Each of the validation squares shows the actual DEM data on the left, and the extrapolated data on the right. The volumetric difference (notably very little) is shown on the squares on the right; loud colors mean loud errors. Steep extrapolations do struggle a bit.

(3) Reaching USGS's figure would require external constraint – drill cores or historical pre-mining topography – which we do not have.

Therefore, step 3 volume for NC000004-006 is best reported as 12.9 M m³ (bracket 12.87–12.99 M), characterized as ~14 % below USGS owing to an undocumented base-definition difference and not as an unresolved error.

At this point it was clear more sites were necessary to tune against. Instead of 10 sites, the algorithm was expanded to 20, across VA, MT, AZ, and NC – and the results were telling. Failures across the band colored a dull picture for the assumption that merely extrapolating the ground from the surroundings is not reliable.

Step Four

With the same resources as before, the project was at a crossroads. Figuring out what was a confounding variable, and what actually caused the issue required a lot of testing. The 20 sites previously determined in step three were formalized in `Step_Four_Trifolds.ipynb` – relative flat surrounding terrain.

The previous process was rerun on all 20 sites in `01_pipeline_and_batch.ipynb`. The same diagnostic poster from the previous step was regenerated for each of the 20 selected sites, as well as validation squares. This would reveal the failures fully: 10 sites with massive, 50% + errors, 6 sites with large errors, and only 4 sites with errors below 20%. The results are in the posters folder. This, coupled with the fact that the validation squares proved volumetrically that pure interpolation was decidedly *not* the issue. However, some loose threads were left behind.

To understand these threads, one must grasp perimeter relief. *Perimeter relief* is the vertical range of the ground around the edge of a waste dump – the difference in elevation between the highest and lowest points along the dump's outline. A dump sitting on a flat plain has low perimeter relief: walk around its base and the ground stays at roughly the same height. A dump filling a valley or draped across a hillside has high perimeter relief: one end of its outline sits far downhill from the other. This number matters because the method estimates the hidden ground *beneath* a dump by looking at the exposed ground *around* it. When the perimeter is flat and even, that surrounding ground is a reliable guide to what lies buried. When the perimeter spans a large drop, the buried surface is more likely to plunge in ways the edges cannot reveal – so perimeter relief is a rough proxy for how trustworthy the buried-ground estimate is.

Another path, a seemingly simple enough plan, existed for us to attempt. Many failing sites sit in valleys, and valleys carry water, so the buried ground beneath those dumps was once threaded with stream channels (drainage channels) that the dump now covers. The hypothesis was that if we could reconstruct where those buried channels ran and how deep they cut, we could rebuild a more realistic buried surface – one with valleys in it – instead of the smooth surface the basic method produces, and close the gap to the USGS volumes.

It was a reasonable idea, and on idealized test cases it worked. But on real sites it made the estimates *worse*, for a fundamental reason: the modern stream, seen in the elevation data, has been diverted *around* the dump, so it no longer marks where the buried channel actually ran. Reconstructing the old channel from the new one meant guessing, and there was no ground truth to check the guess against. The conclusion was not "drainage doesn't matter" but "the buried channel cannot be recovered from the surface alone." So the hypothesis was documented as an instructive dead end rather than a fix – it sharpened *why* the problem is hard, which is itself a result.

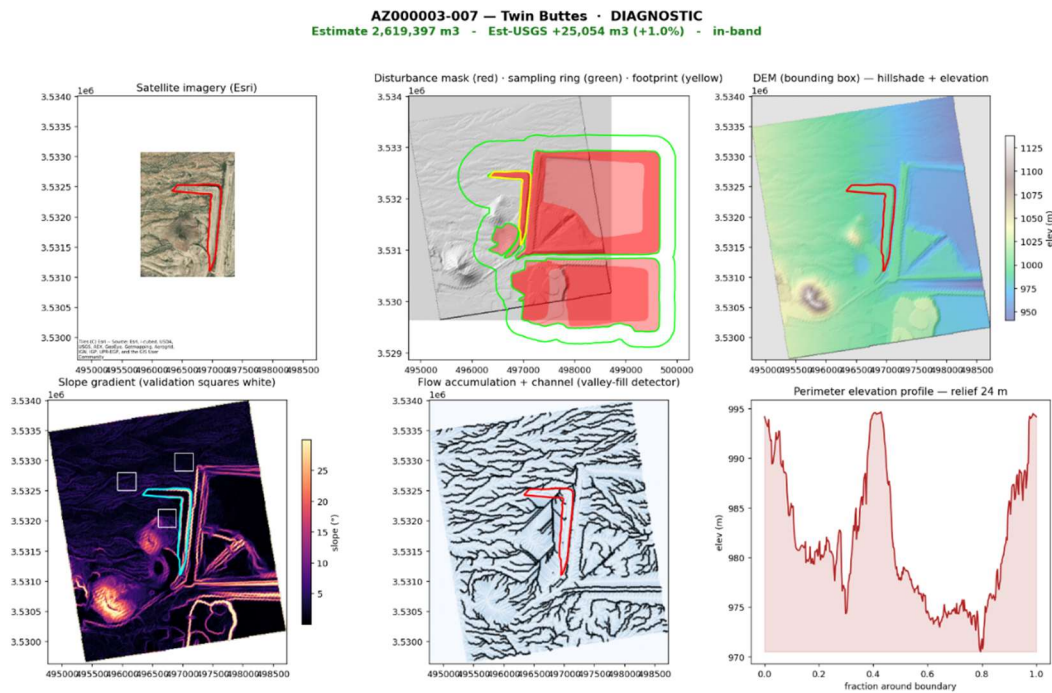


Figure 7: Drainage diagnostic – this is one of the most successful entries in the drainage hypothesis. The channel relief, the slope gradient, and the DEM's worked in its favor.

VA000003-002 — Austinville Zinc and Lead Mines · DIAGNOSTIC
 Estimate 194,808 m³ - Est-USGS -1,555,441 m³ (-88.9%) - FAILED

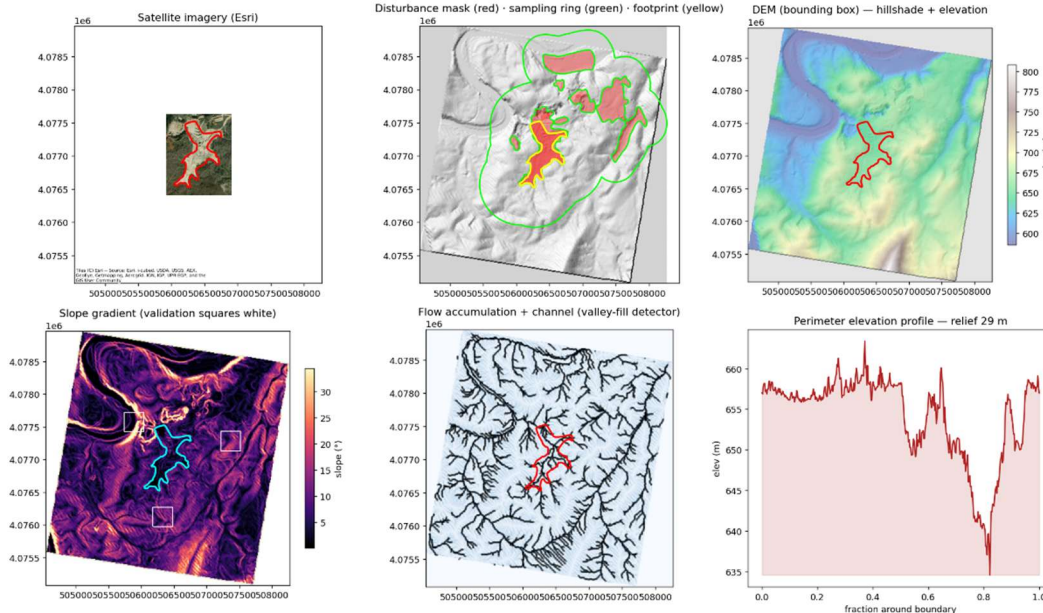


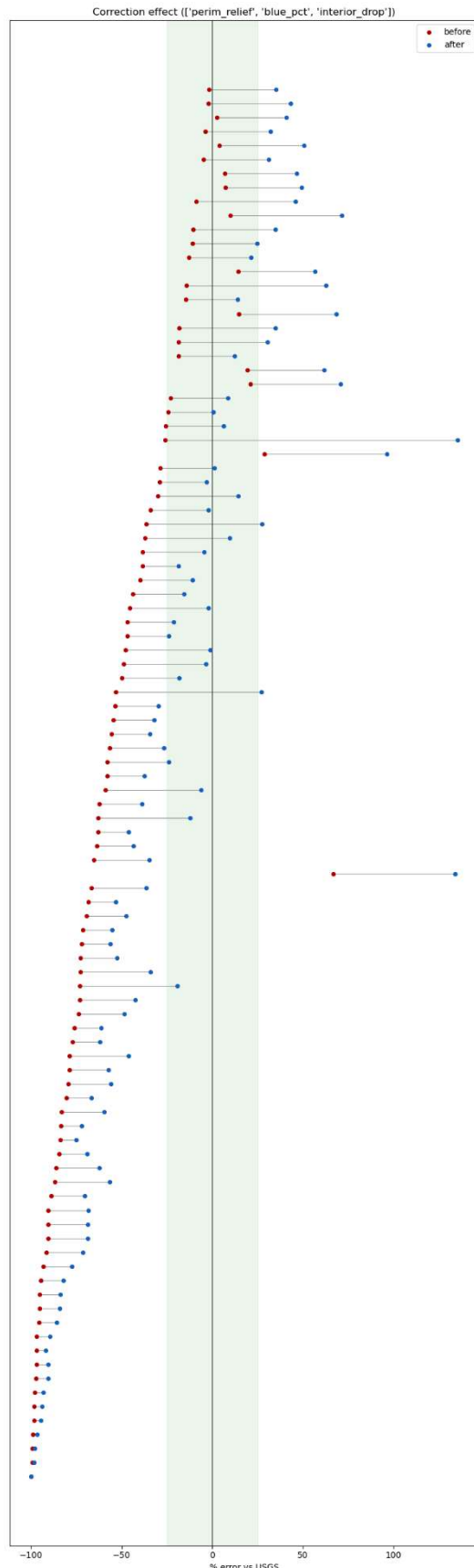
Figure 8: Drainage diagnostic – this is an example of the many failures in the drainage hypothesis. The channel relief did not work, evidently.

In O2_drainage.ipynb, the thread is formalized. The two worst failures rerun with a cross section of the site, revealing that the failures are not a simple fix with regards to interpolation, or even the drainage hypothesis. Step four ultimately ends in a dead end for progress, but it is intensely instructive: the problem is not easily solved with satellite imaging, and a surface-level hypothesis like drainage channels aren't going to be a silver bullet. Sites are dramatically different from location to location, and the buried surface they are built upon is simply impossible to guess. However, more threads remain.

Step Four – V2

The reason this is not a step five is because nothing *new* in the manner of progress is made. Instead, the confounding variables are isolated and identified – the limits of the project are revealed here. The simplest approach first was to attempt a basic regression: perhaps the high median per-site error could be reduced.

A *regression* is a statistical tool for finding whether one quantity can be predicted from others. Here, the quantity we wanted to predict was the *error* of our volume estimate – how far below the USGS figure it fell – and the predictors were features we can measure directly from the elevation data, such as perimeter relief and how much of the dump the method visibly struggles with. A well-formulated model cut the typical per-site error roughly



in half and, more importantly, removed almost all of the *aggregate* bias – when the volumes of many sites are summed into a regional total, the individual over- and under-estimates largely cancel, and the corrected total lands within a few percent of the USGS sum.

A first %-error model on the 100-site (see *Figure 9: A visualization of the regression's result*) set lowered the median from ~59–65% to 46%; a better-formulated log-ratio model – $\ln(\text{USGS}/\text{estimate})$ on perimeter relief and the 'blue' clip fraction – reached $R^2 = 0.86$ and a median of ~34%. Crucially, at the basin scale the correction removed almost all aggregate bias: the raw pipeline under-counts total regional waste by roughly half (–58%), and the corrected total lands within –4 to –6% of the USGS sum – with the caveat that a near-zero net arises partly from opposing per-site errors cancelling, a cancellation specific to this 100-site sample and not bounded by a confidence interval here. Per-site it remains a bias-reducer; the regional total is where it delivers.

The regression made the confounding structure explicit. Perimeter relief alone carried no predictive signal ($R^2 = 0.00$) – large, high-relief plateau dumps in Arizona sit in the error distribution right alongside low-relief valley fills in Virginia, so relief by itself cannot separate the two. Only when combined with the 'blue' clip fraction (a measure of how much the reconstructed base pokes above the current surface) did a usable model emerge. Adding further geometric predictors – an interior-drop 'buried-valley' signature, footprint elongation, footprint area – moved the leave-one-out error negligibly while in-sample R^2 crept upward, the textbook signature of a model that has reached its ceiling.

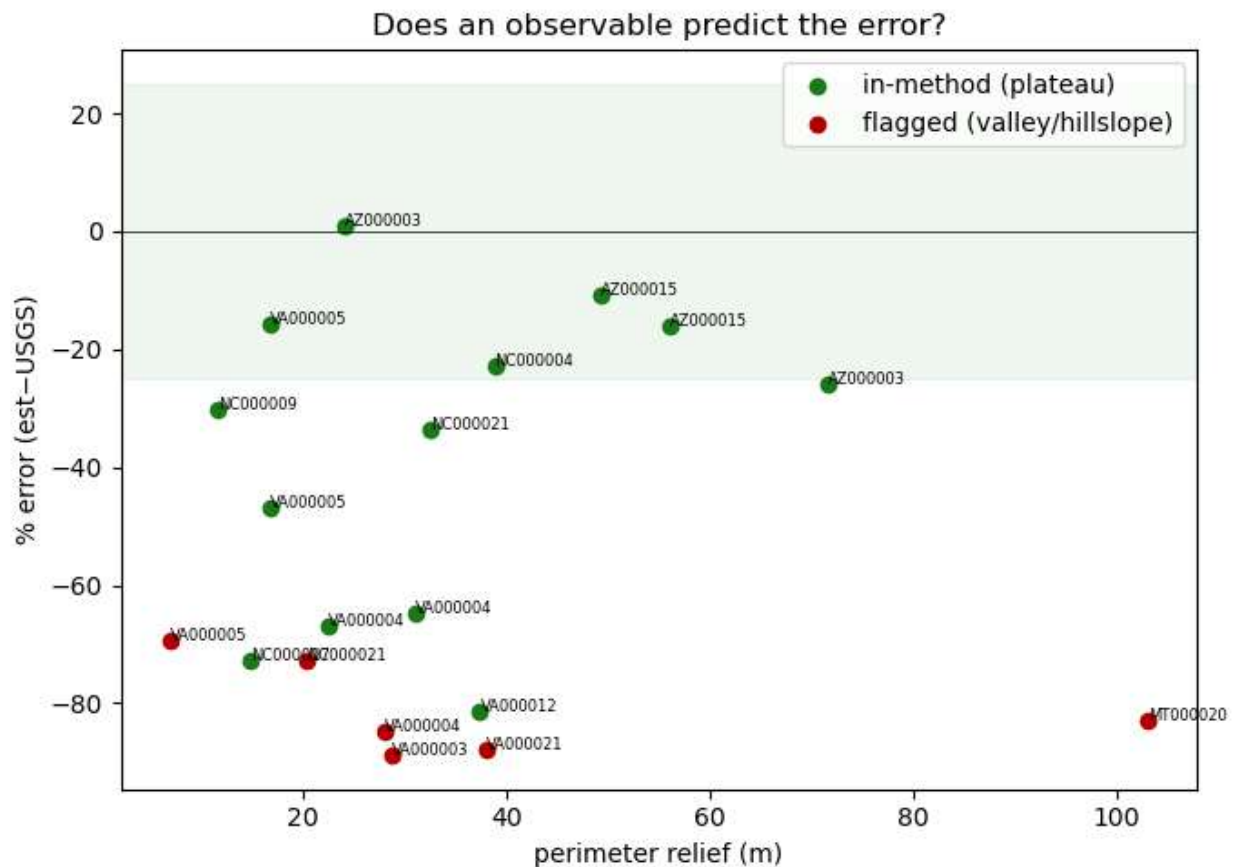


Figure 10: Graph visualization of the regression; this time revealing that the regression is not related to the perimeter relief of the site.

Scaling the sample from 20 to 100 sites (04_scale_100.ipynb), stratified by state and volume decile and restricted to raised-fill material types, was decisive. At $n=100$, with proper cross-validation, ~46% (%-error form) and ~34% (log-ratio form) are the honest per-site numbers, and they do not fall further no matter what is added. The floor is real. Surface observables can proxy the buried deficit well enough to halve the aggregate bias, but cannot recover per-site volume, because the missing information is not on the surface.

Accepting External Truth (Control Points)

Ultimately, the previous steps simply could not capture the scale of the problem on their own. However, contact with an actual USGS office revealed something anticipated early on: USGS's own volumes are not derived from surrounding terrain but from field work – the Tungsten Queen figure, for instance, rested on a georeferenced pre-mining

topographic map and drill cores that fixed the true depth to contact. A modern DEM cannot manufacture any of that. What the pipeline can do is accept it when it exists.

This is exactly what 05_control_points.ipynb is built to do. The interpolation, which normally samples only the natural ground ringing the footprint, is extended to fold in either digitized contours from an old topographic map, or drill cores. These are added to the control set alongside the natural ground. On a synthetic valley fill with known ground truth, folding real interior control moved the estimate from -13% to $+11\%$ – the mechanism reshapes the base as intended. No real data was available during the project, so this stands as validated infrastructure: the pipeline is now field-data-ready, matching USGS wherever such data is supplied and defaulting to its estimate everywhere else.

The Resolution Question – Does 1 m DEM Help?

A standing suspicion, and an explicit advisor question, was whether the coarse 10 m DEM was itself the source of the error – sharper toe, edges, and slope at 1 m might refine the estimate toward USGS. Duke’s network blocks the 1 m elevation service (it is simply too much to download at once), so the data was fetched manually through Google Colab and the pipeline run offline against the local tiles in 06_dem_1m_test.ipynb.

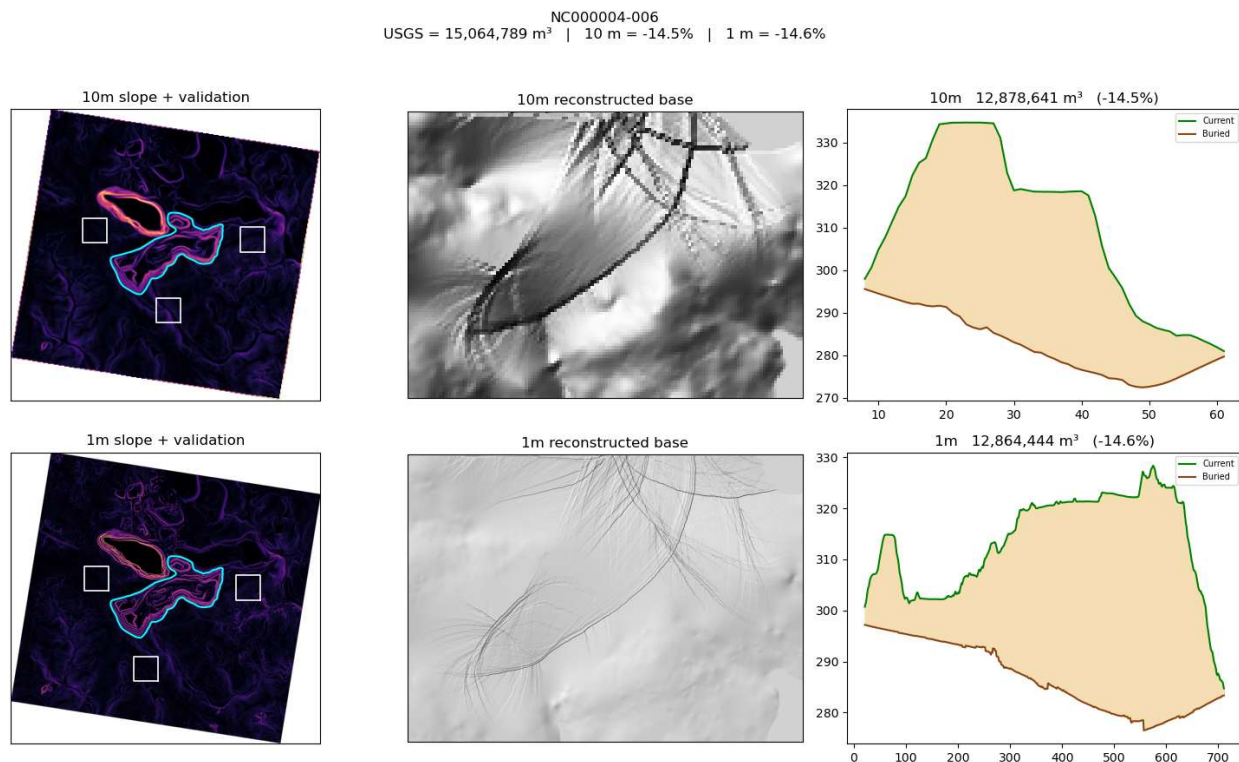


Figure 11: Mission Complex with 10 meter DEM above and 1 meter below.

NC000004-006
USGS = 15,064,789 m³ | 10 m = -14.5% | 1 m = -14.6%

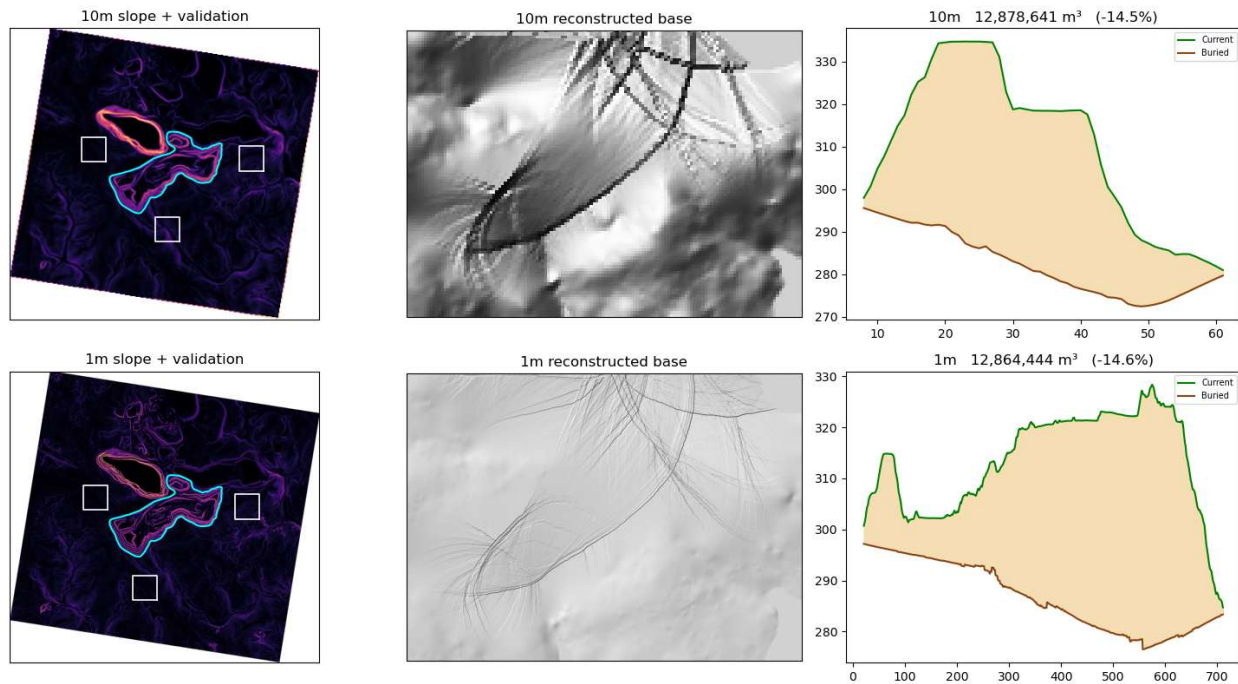


Figure 12: Hallman-Beam with 10 meter DEM above and 1 meter below.

On the two anchor sites the answer was unambiguous: Mission Complex went from -14% at 10 m to -15% at 1 m; Hallman-Beam held at -15% both ways. A hundred-fold increase in resolution changed the reconstructed base and the volume negligibly. Resolution is not the bottleneck.

Two sites cannot carry that claim, so the test was scaled to the ten worst-error sites in 07_dem_1m_worst10.ipynb. Across those ten, the 1 m-vs-10 m volume shift was under 1% on nine of them – confirming at scale that the 80–120% errors are buried-surface, not resolution. The validation-square reconstructions, saved to the Proof folder, make this visible: even with a footprint resolved a hundred times more finely, the volume barely moves, because the spline is the same, and the buried deficit lies beneath it.

One site broke the pattern instructively. VA000016-003 improved from -80% at 10 m to -42% at 1 meter. Viewed as the only merit for this method so far, the hypothesis was tested properly in 08_footprint_resolution.ipynb.

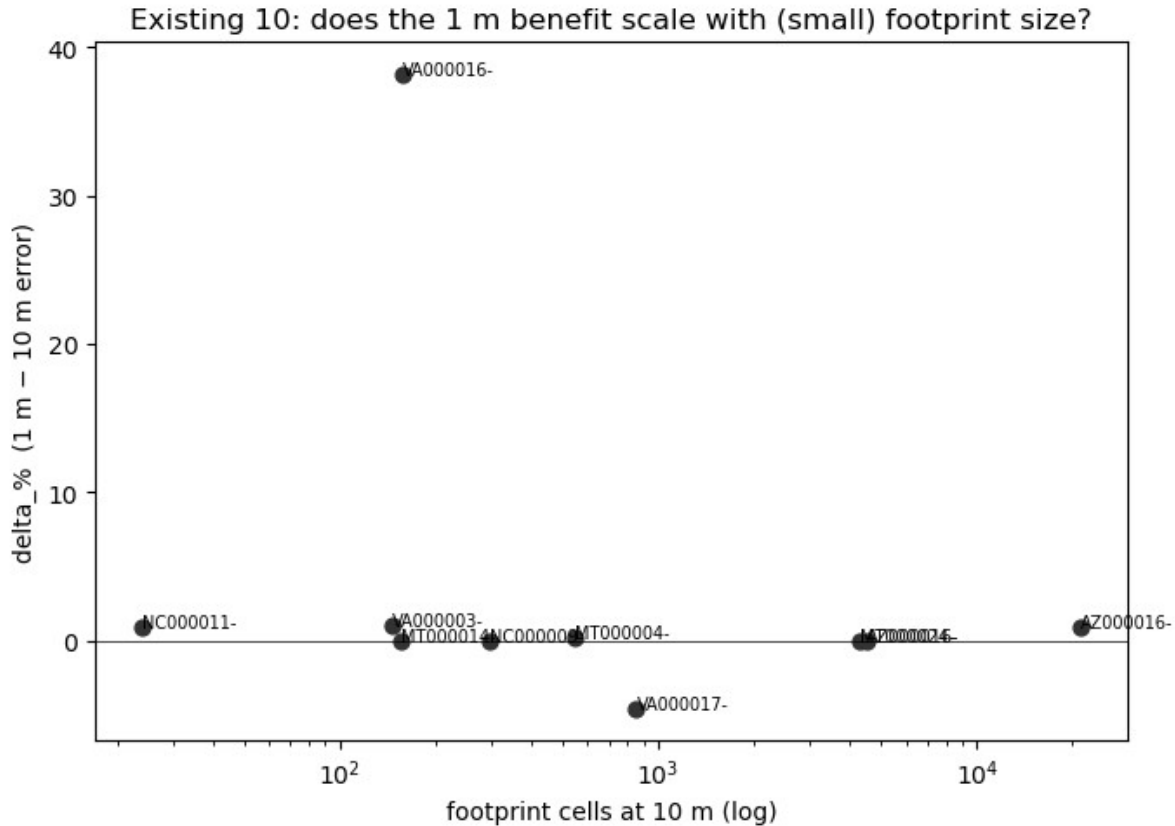


Figure 13: All hope appeared lost for the DEM resolution hypothesis – then the Virginia site jumped in accuracy by 38 points.

1 m helps specifically when the footprint is small relative to the 10 m cell, because at 10 m a small dump is only tens of cells and its toe and edges are truncated. A targeted sample of small-footprint dumps plus medium-footprint controls, each fetched at native 10 m and 1 m to avoid any resampling artifact, confirmed it: the small group improved by a median of +11.9 points (9 of 10 sites), while the controls moved +0.2 points (0 of 6).

The benefit is real, footprint-size-specific, and partial: most small sites remain badly under-counted at 1 m, so resolution was a component of their error, never the whole of it. This suggests a practical rule – apply 1 m selectively to footprints under roughly 100 cells at 10 m, and keep 10 m elsewhere – though it rests on a small sample (ten small sites, two of which over-corrected), so it is offered as a tested tendency rather than a settled threshold.

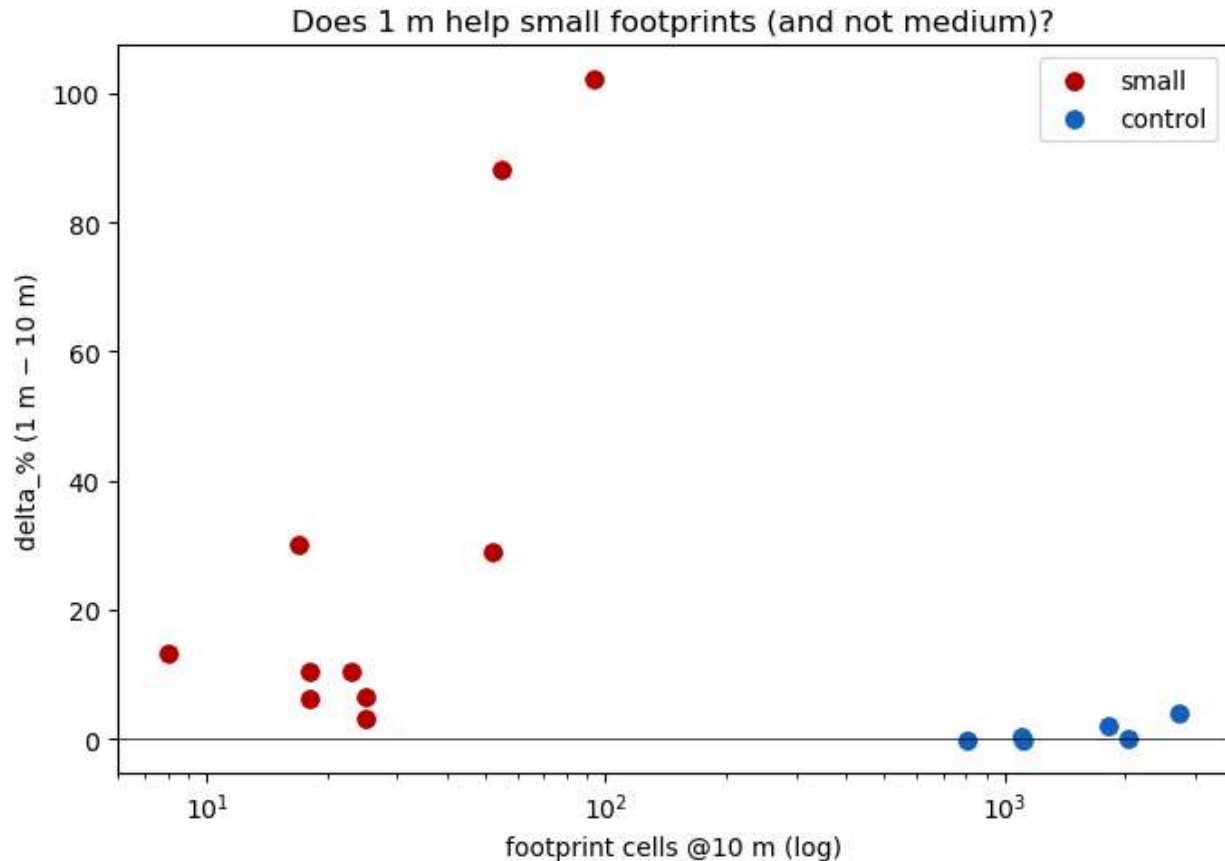


Figure 14: A visualization of if 1 meter resolution actually improves the volume estimate.

Is the Spline the Problem? (Kernel Comparison)

A final loose thread, and the second advisor question, was whether thin-plate spline was an arbitrary choice – whether a different interpolation kernel might be more consistent. 09_spline_kernels.ipynb refits the identical control points on the identical grid with five kernels (thin-plate, linear, cubic, multiquadric, gaussian) across the 20 selected sites at 10 m and the 18 with 1 m data, comparing every kernel's volume to both USGS and to thin-plate.

The four stable kernels agree with thin-plate at the median to the rounding. A 0.0% typical shift. So kernel choice does not drive the estimate; thin-plate is the consensus surface, not an arbitrary one. No kernel improves on it: median error stays pinned near 57% regardless, because the residual is buried-surface and invisible to any interpolator. Gaussian proved unstable and was rejected: on one site it overshoot to +496% relative to thin-plate, a spurious deep dish rather than a better fit. The validation squares were verified

volumetrically per kernel and the full results saved to Proof/kernel_comparison.csv for the record.

A Note on Interpretation

Two framing points govern how the numbers in this report should be read. First, every error figure is measured against the USGS calculated volumes, which are the only authoritative reference available – yet those volumes are themselves of undocumented method and are not independently validated. Our ‘errors’ therefore quantify agreement with USGS, not accuracy in an absolute sense. Where our estimate and USGS disagree, the discrepancy could originate in our method, in the USGS figure, or in both; without external ground truth neither can be certified, and where we attribute a gap to a specific cause that attribution is an inference by elimination rather than a measured result.

Second, this report does not claim to have solved the volume problem, nor does it treat the method’s limits as failures. It reports what a surface-only approach can and cannot do, holds open the possibility that untested levers – the spline’s smoothing parameter, the base-clipping guard, or sites in regions beyond the four sampled states – could still move individual results, and presents its central negative finding as a bounded, testable claim rather than a certainty.

Conclusion:

The project set out to tune a terrain-based volume estimator toward USGS. What it produced is more honest and more useful than tuning: a fully characterized method with a stated domain of validity, and a rigorous account of exactly where and why it fails.

The method works for flat-lying and plateau dumps, where the perimeter approximates pre-mining ground and the residual is a modest base-definition offset. It fails, structurally, on valley and hillslope fills, where the buried valley floor plunges below the perimeter the spline is anchored to – a void surface terrain cannot see. That failure is not a bug to be fixed with a better spline, a finer DEM, a drainage heuristic, or more training sites; each of those was tried and each hit the same wall.

That is the central result, and its strength is in the convergence. The ~50% per-site median error proved stable across sample size (19→100), error model (%-error vs log-ratio), DEM resolution (10 m vs 1 m), spline kernel (five tested), and to gating strategy. Five lines of attack converging on the same floor. These lines are not fully independent – they share the same footprints, DEM source, reconstruction method, and benchmark – so the

convergence is strongly consistent with, rather than proof of, a physical limit. The most likely such limit is missing buried-surface information, exactly the field data (pre-mining topography, drill cores) from which USGS derived its own numbers.

The deliverables reflect this split honestly. At the regional or basin scale, the log-ratio correction reduces aggregate bias from -58% to within -4 to -6% of the USGS total: a usable answer to 'how much waste is in a region.' Prospective extractors can reasonably muse that within a large enough region, the value estimated from this program will actually be yielded from the earth.

At the per-site scale, volumes are reported as brackets, not point estimates, with the domain-of-validity diagnostic flagging which sites the method can and cannot measure. And the control-point hook stands ready to cross the floor the moment field data exists, matching USGS where truth is available and falling back to a characterized estimate everywhere else. The information gap the project set out to fill is, for adequately-resolved raised-fill dumps at regional scale, meaningfully narrowed – and for the valley fills that resist, precisely explained.

The Future: What Remains

This project reached a natural and well-defined stopping point: every avenue that could be tested with the available data was tested, and the limits of a surface-only method were established rather than merely encountered. What remains falls into three categories. One project is immediately actionable, one of substantial research, and one deliberately out of scope.

I. Sourcing real buried-surface data (the highest-value next step). The single result that would most advance this work is not more code – it is one real dataset. The control-point pipeline is built and validated but has never been fed genuine field data, because none was available during the project. A single pre-mining topographic map digitized into elevation contours, or a set of drill-core logs giving depth-to-original-ground, for even one valley-fill site would do two things at once: it would let the pipeline produce a genuinely measured volume for that site, and it would provide the ground truth needed to validate the entire buried-surface hypothesis directly rather than by inference.

II. Reproducing an independent footprint classification. The project trusted the mine-feature outlines provided in the national inventory. An open question is whether those outlines are themselves a source of error. Something like a mis-drawn boundary, or a polygon that encloses non-waste ground, would corrupt a volume estimate regardless of

how good the terrain reconstruction is. I believe this is the likely cause of the sites that *over-estimate* rather than under-estimate. Independently classifying each dump's footprint from aerial imagery, and validating its boundary against sharp breaks in the elevation slope, would test this.

III. Machine-learning terrain reconstruction. A more ambitious idea is to train a neural network – a vision transformer or a segmentation model – to reconstruct buried surfaces directly. This was set aside, and the project's own findings explain why. The limiting factor is the interpolation but the *absence of information* about the buried surface in the input data. A model trained only on surface elevation faces the same wall every method in this project hit; it could only succeed if trained on sites where the true buried surface is known. The most sensible sequencing is therefore to acquire buried-surface datasets first, and only then revisit machine-learning reconstruction as a way to generalize from them.

Returning to the valuation this project set out to serve: of the four terms in *Value = Volume × Grade × Price × Recovery*, this work addresses only Volume, and characterizes exactly how far that term can be trusted. At the regional scale the corrected volume lands within a few percent of USGS, so a region-wide resource-and-value estimate is well supported: multiply the basin volume by an average grade and price and the answer is meaningful. At the single-site scale the volume is a bracket, not a point, and for valley/hillslope fills it is a bounded underestimate — so a per-site valuation built on it inherits that same uncertainty and should carry it forward explicitly. The volume term is therefore *usable for prospecting triage at regional scale, and honest about its own error at the site scale* — which is precisely the input a downstream grade-and-price model needs, provided it is told which regime each number came from.

In short, the work is not unfinished so much as *bounded*: it has done what surface data permits, characterized precisely where that runs out, and built the mechanism to continue the moment the missing data becomes available.